HONG KONG EXAMINATION AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2026

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Candidate Number									

MATHEMATICS Compulsory Part
PAPER 1
Question-Answer Book

8:30 am – 10:45 am (2¼ hours)

This paper must be answered in English

INSTRUCTIONS

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7, 9 and 11.

(2) This paper consists of THREE sections, A(1), A(2) and B.

(3) Attempt ALL questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

(4) Graph paper and supplementary answer sheets will be supplied on request. Write your Candidate Number, mark the question number box and stick a barcode label on each sheet, and fasten them with string INSIDE this book.

(5) Unless otherwise specified, all working must be clearly shown.

(6) Unless otherwise specified, numerical answers should be either exact or correct to 3 significant figures.

(7) The diagrams in this paper are not necessarily drawn to scale.

(8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

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SECTION A(1) (35 marks)

1. Make m the subject of the formula $\frac{m}{2}-1=\frac{5m-n}{3}$. (3 marks)

2. Simplify $\frac{(uv^2)^5}{u^7v^{-3}}$ and express your answer with positive indices. (3 marks)

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3. The coordinates of the points P and Q are $(-6,5)$ and $(2,1)$ respectively. P is translated downwards by r units to R.

   (a) Write down, in terms of r, the coordinates of R.

   (b) Find r such that $\angle PQR = 90°$.

(3 marks)

4. Factorize

   (a) $5x^2y + 17xy - 12y$,

   (b) $5x^2y + 17xy - 12y - 10x + 6$.

(4 marks)

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5. The sum of the number of blue pens and the number of red pens in a box is 795. If 123 blue pens are taken out from the box, then the number of blue pens in the box is 25% less than the number of red pens in the box. Find the number of red pens in the box. (4 marks)

6. Consider the compound inequality

$$3x-4<5\leq\frac{2-7x}{6} \quad \ldots\ldots\ldots\ldots\ldots \ (*) \ .$$

(a) Solve (*).

(b) Write down the greatest integer satisfying (*).

(4 marks)

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7. A profit of $162 is made by selling a ring at a discount of 15% on its marked price. The marked price of the ring is $270 above its cost. Find the percentage profit. (4 marks)

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8. In Figure 1, ABC is a triangle. D is a point lying on AB. Let E and F be points lying on CB produced and CD produced respectively such that $BD \,//\, EF$ and $BC = EF$. It is given that $BD = 8$ cm and $BE = 6$ cm.

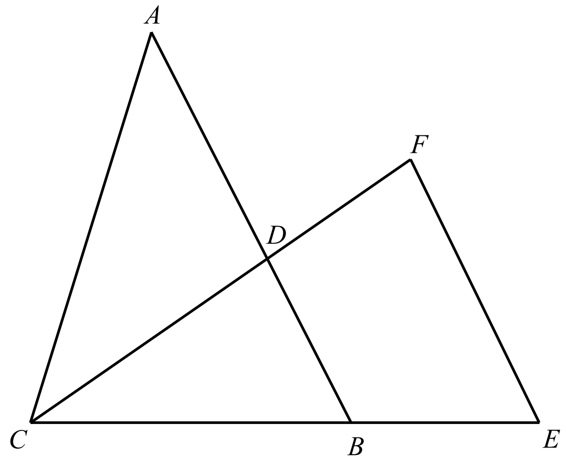

Figure 1

(a) By considering a pair of similar triangles, find the length of BC.

(b) Suppose that $AD = 10$ cm. Prove that $\Delta ABC \cong \Delta CEF$.

(5 marks)

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9. The frequency distribution table and the cumulative frequency distribution table below show the distribution of the weight of 20 apples.

Weight (grams)	Frequency
91 – 100	2
101 – 110	a
111 – 120	6
121 – 130	7
131 – 140	h

Weight less than (grams)	Cumulative Frequency
100.5	2
110.5	6
120.5	12
130.5	b
140.5	k

(a) Find a and b.

(b) Find the mean of the distribution.

(5 marks)

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Section A(2) (35 marks)

10. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is $7:3$.

(a) Find w. (3 marks)

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(3 marks)

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11. It is given that $f(x)$ partly varies as x^2 and is partly constant. Suppose that $f(4)=0$ and $f(8)=-9$. Denote the graph of $y=f(x)$ by Γ.

(a) Find the y-intercept of Γ. (3 marks)

(b) Γ cuts the x-axis at the points U and V while Γ cuts the y-axis at the point W. Find the perimeter of ΔUVW. (3 marks)

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12. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L. (2 marks)

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(5 marks)

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13. A solid metal sphere of radius 10 cm is melted and recast into three similar solid right circular cylinders *X*, *Y* and *Z*. Let r_1 cm, r_2 cm and r_3 cm be the base of radii of *X*, *Y* and *Z* respectively. It is given that $r_1 : r_2 : r_3 = 2 : 3 : 5$.

(a) Express the volume of *Y* in terms of π. (3 marks)

(b) A craftsman finds that the height of *Y* is 9 cm.

(i) Find r_2.

(ii) The craftsman claims that the total surface area of *Z* is greater than the surface area of the sphere. Do you agree? Explain your answer.

(5 marks)

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14. When the polynomial $g(x)$ is divided by x^2+2x+a, the quotient and the remainder are $x+13$ and $b(x-3)$ respectively, where a and b are constants. It is given that $g(x)$ is divisible by $x-3$.

(a) Find a. (3 marks)

(b) Is there only one possible value of b such that at least two roots of the equation $g(x)=0$ are equal? Explain your answer. (5 marks)

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Section B (35 marks)

15. A bag contains 3 red balls, 3 white balls and 5 black balls. If 4 balls are randomly drawn from the bag at the same time, find

(a) the probability that exactly 2 red balls are drawn, (2 marks)

(b) the probability that exactly 2 red balls or exactly 2 white balls are drawn. (2 marks)

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16. It is given that P, Q, R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C. (3 marks)

(b) The tangent of C at P and the tangent to C at R intersect at the point H. Is ΔHPR a right-angled triangle? Explain your answer. (3 marks)

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17. In Figure 2, the base of the pyramid *VABCD* is the quadrilateral *ABCD*. The face *VAB* of the pyramid lies on the horizontal ground. It is given that $AB = 90$ cm, $AD = 70$ cm, $CD = 100$ cm, $\angle ADC = 55°$ and $\angle BCD = 130°$.

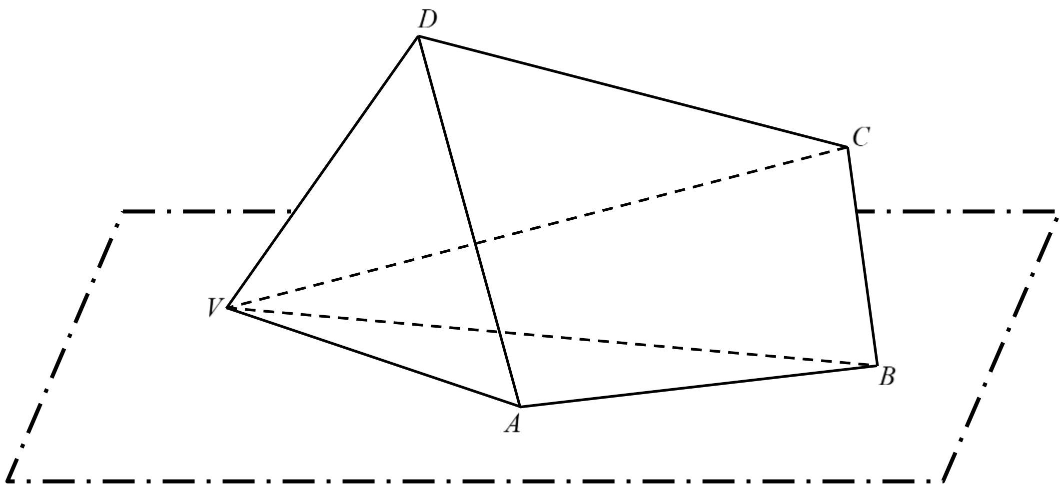

Figure 2

(a) Find

(i) the distance between *A* and *C*,

(ii) $\angle BAD$.

(4 marks)

(b) It is given that the $VD = 80$ cm. A student finds that the angle between the plane *ABCD* and the horizontal ground $65°$. The student claims that the angle between *VD* and the horizontal ground is less than $50°$. Is the claim correct? Explain your answer. (2 marks)

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18. Let $\mathrm{G}(n)$ be the nth term of a geometric sequence. It is given that $\mathrm{G}(2)=12$ and $\mathrm{G}(5)=768$.

(a) Find $\mathrm{G}(1)$. (3 marks)

(b) Denote $\mathrm{T}(n)=\log_2 \mathrm{G}(n)$ for all positive integers n. Find

(i) $\mathrm{T}(n+1)-\mathrm{T}(n)$,

(ii) the greatest value of k such that $\mathrm{T}(1)+\mathrm{T}(2)+\mathrm{T}(3)+\ldots+\mathrm{T}(k)<10^6$.

(4 marks)

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19. Let $\mathrm{h}(x)=-x^2+30kx+4x-225k^2-40k+3$, where k is a positive constant. On the same rectangular coordinates plane, denote the vertex of the graph of $y=\mathrm{h}(x)$ and the vertex of the graph of $y=14-\mathrm{h}(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A. (2 marks)

(b) Write down, in terms of k, the coordinates of B. (1 mark)

(c) The coordinates of point C are $(2+48k, 7-36k)$. Let I and J be the in-centre and circumcentre of ΔABC respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6-22k, 6-15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(9 marks)

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END OF PAPER

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